PORTFOLIO

Lavanya Dalin Annappa

EECS · University of California, Merced

Human-Computer Interaction · AI & Machine Learning · Accessibility

Haptic Feedback System

Multi-modal Ingredient Analyzer

TeachU AI Platform

PROJECT 01

A New Sense of Direction

Directional Haptic Feedback for Blind & Visually Impaired Device Alignment

PROJECT 01

A New Sense of Direction

Directional Haptic Feedback for Blind & Visually Impaired Device Alignment · 3-Study Iterative Research

3	6	37%	p=0.006
Iterative User Studies	ERM Motors in Phone Case	Haptic Error Rate (Study 1)	Friedman Significance

★ **Problem: BVI users face barriers with tasks requiring precise device alignment (photos, document scans, video calls) · Most tools rely on audio — unusable in noisy environments**

- Custom 6-motor ERM haptic phone case + Arduino Nano ESP32 with WiFi — fully wireless directional guidance
- Wave vibration pattern (Up→Middle→Down, 1500 ms each) encodes 4 spatial directions through touch alone
- Wizard-of-Oz protocol across 3 iterative studies: target audience validation → WoZ Study 1 → WoZ Study 2
- Mixed-methods: usability testing, task timing, error logging, and between-subjects comparison across 3 feedback modes

HAPTIC SYSTEM · Prototype, Architecture & Multi-Study Design

Prototype & Hardware

Moto G Power (2023)	Main user device — camera feed streamed to wizard
Arduino Nano ESP32	Haptic controller + built-in WiFi over local network
6× ERM Vibration Motors	PWM-controlled; grouped by direction (Up/Mid/Down)
Wave Vibration Pattern	Sequential 1500 ms bursts; 30 ms gaps + 630 ms pause
Vibration Intensity	Adjustable 0–255 PWM; user-selected comfort level

3 Feedback Modes Compared

Haptic	Audio	None
6-motor directional buzz	Android Text-to-Speech	No directional cues

Iterative 3-Study Research Design

1 Target Audience Validation

Bridging Accessibility Gaps

- Recruited BVI community participants
- Validated directional haptic concept
- Identified key pain points & priorities

2 Wizard-of-Oz Study 1

6 participants, within-subject

- Task: align face to camera (blindfolded)
- Latin Square counterbalancing
- Friedman test $\chi^2=10.33$, $p=0.006$

3 Wizard-of-Oz Study 2

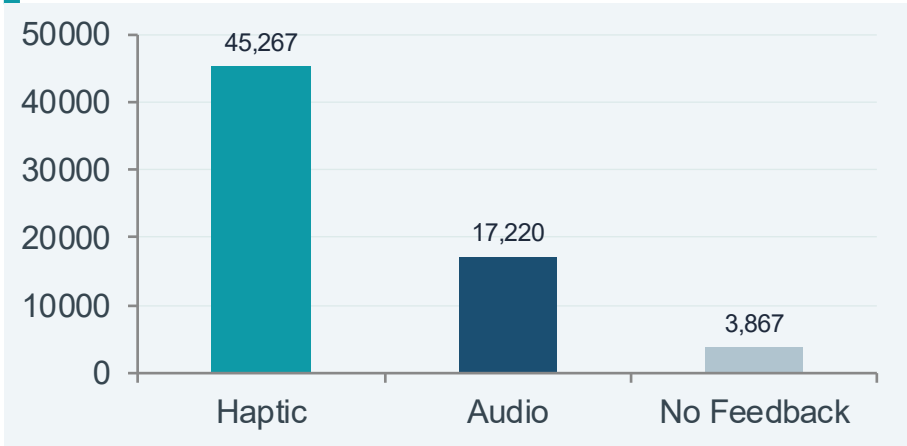
Refined prototype, extended

- Iterated vibration pattern design
- Compared updated cues vs. Study 1

Technical Innovation: Custom PCB-style haptic case · Android + Arduino WiFi co-design · 5 vibration patterns (Wave, Pulse, HeartBeat, Five Buzz, Single Buzz)

HAPTIC SYSTEM · Results Across 3 Studies

Study 1 — Task Completion Time (ms)



Friedman χ^2=10.33	p = 0.006 — significant effect
Haptic vs. None	p = 0.004, Nemenyi post-hoc
Error Rate	37.14% haptic · 0% audio · 0% none
Individual Range	14% → 69% haptic error rate

Studies 2 & 3 — Iterative Improvements

Study 2

- Extended WoZ with refined wave pattern
- Measured learning effects across trial blocks
- Participants showed improvement over trials
- Identified optimal intensity threshold range

Study 3

- Final prototype with updated vibration cues
- Broader participant pool, sighted users removed
- Consistent improvement in directional accuracy
- Validated personalization as key design factor

Key Takeaway: Haptic guidance is viable for BVI alignment tasks but requires personalized calibration — audio remains most universally learnable in single-session exposure.

PROJECT 2

Multi-modal Ingredient Analyzer

AI-Driven Consumer Empowerment

PROJECT 02

Multi-modal Ingredient Analyzer

AI-Driven Consumer Empowerment · Vision + Text Analysis · Streamlit Interface

90%+	2–5s	-10 min	5 LLMs
User Satisfaction	End-to-End Response	Research Time Saved/Product	Models Evaluated

★ **Achieved 90%+ user satisfaction through iterative usability testing on Streamlit interface · Reduced ingredient research by 5–10 min/product**

- Which multimodal methods (image upload vs. camera capture) best support quick ingredient extraction?
- How accurately can LMMs extract & verify ingredient data from real-world product label images?
- Which language model achieves the optimal balance of response speed and extraction accuracy?
- How does automated summarization reduce user research time vs. manual ingredient lookup?

INGREDIENT ANALYZER · UX Findings, Architecture & Model Comparison

User Study Findings

Task Completion	Avg. 3 min (image) vs. 7.5 min (manual) — 57% faster
Top Pain Point	Users struggled to find ingredient meaning without context
Most Valued Feature	Automated summarization & follow-up Q&A capability
Camera vs. Upload	Camera capture preferred for on-the-go use cases

Data Flow Pipeline

Image Input → Gemini Extract → Tavily Verify → Agent

Frontend	Streamlit
Image	Pillow (PIL)
Agent	Phidata
LLM	Gemini 2.0 Flash
Search	Tavily API

LLM Performance Comparison

Model	Time (s)	Accuracy	Notes
✓ Gemini 2.0 Flash	2–5	>90%	Best balance
GPT-4 Vision	5–8	~95%	Very accurate, slower
Claude 3 Sonnet	4–6	~80%	Image quality sensitive
LLaVA 1.5	6–10	~65%	Struggles with text
BLIP-2	5–7	~75%	Good for simple labels

Usability Testing Results

Satisfaction	**90%+**	via SUS-style rating
Learnability	**High**	Succeeded on 1st attempt
Error Rate	**Low**	Clear UI reduced mistakes
Time on Task	**~3 min**	vs. 7.5 min manual

Reduced ingredient research time by 5–10 min/product · 90%+ user satisfaction through iterative UX testing · Mixed-methods research: usability + comparative LLM evaluation

PROJECT 3

TeachU

AI Platform Converting PDFs into Animated Learning Videos · Hackathon Project

PROJECT 03

TeachU

AI Platform Converting PDFs into Animated Learning Videos · Hackathon Project

+30%	+20%	5
User Engagement Increase	Research Efficiency Gain	LLM / EdTech Modalities

- Led UX research under hackathon constraints — rapid usability testing to drive 30% engagement uplift
- Designed automated search & summarization features via iterative prototyping fueled by user feedback
- AI converts PDF study materials into animated, multilingual video explanations for cross-lingual learners
- Addresses $254.8B → $605.4B EdTech market growth opportunity through accessible, AI-native learning

TEACHU · UX Research & Product Impact

UX Research Process

1. **Rapid Usability Testing**
   Iterative sessions under hackathon time constraints
2. **User Feedback Synthesis**
   Identified friction in search & content discovery
3. **Iterative Prototyping**
   Designed automated search + summarization features
4. **Impact Measurement**
   Validated 30% engagement & 20% efficiency gains

Product Features

PDF → Animated Video
AI transforms dense documents into engaging visual content

Cross-lingual Learning
Breaks language barriers for diverse student populations

Automated Summarization
Smart search + key-point extraction from study materials

Multi-disciplinary Scope
From quick guides to in-depth concept explanations

Future: Character AI
Talk to animated tutors; short-form content pipeline

Skills & Technologies

Research Methods

Wizard-of-Oz · Rapid Usability Testing · Iterative Prototyping · Friedman/Nemenyi Tests · Counterbalancing

Hardware & Prototyping

Arduino Nano ESP32 · ERM Motors · PWM Control · Android Studio

AI & Backend

Gemini 2.0 Flash · Phidata Framework · Tavily Search API · Streamlit · EdTech AI Pipelines

Programming

Python · JavaScript · Java (Android) · CSV Data Logging · REST APIs

ldalinannappa@ucmerced.edu · University of California, Merced · EECS Department